Having *fun* with RISC-V: Data Annex

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TABLE I: Full breakdown behind Table II of the letter. Compute window (the measurement loop), both compilers. IPC is instret/cycles. Ratios are GHC/MicroHs; *whole* is the whole-run cycle ratio including RTS start-up.

Program	MicroHs (RV32-xfun)					GHC (RV32)			GHC/MicroHs		
	cyc	instret	IPC	combi	fn.y	cyc	instret	IPC	cyc	instret	whole
flite											
Permsort	36 051	27 142	0.75	11 758	321	697 202	545 378	0.78	19.34	20.09	34.45
Queens2	49 257	38 521	0.78	15 162	1 121	787 836	634 362	0.81	15.99	16.47	33.28
Taut	103 472	82 268	0.80	29 945	1 382	1 354 694	1 079 880	0.80	13.09	13.13	30.12
Ordlist	32 764	24 526	0.75	13 488	419	385 074	309 580	0.80	11.75	12.62	33.77
Braun	170 346	113 383	0.67	54 570	260	1 978 346	1 626 875	0.82	11.61	14.35	25.35
Sumpuz	28 192	20 902	0.74	6 616	650	304 720	235 986	0.77	10.81	11.29	34.13
Sudoku	1 426 304	920 354	0.65	250 579	9 337	12 519 991	10 626 218	0.85	8.78	11.55	12.35
TreeSum	620 772	417 892	0.67	114 681	24 573	4 270 049	3 608 820	0.85	6.88	8.64	14.44
TreePari	108 292	85 839	0.79	30 449	1 024	720 013	608 424	0.85	6.65	7.09	26.24
Adjoxo	64 202	46 064	0.72	18 921	341	399 772	321 945	0.81	6.23	6.99	31.04
Mss	670 984	569 051	0.85	70 794	12 430	3 484 015	2 992 377	0.86	5.19	5.26	12.59
Knuthbendix	81 353	54 625	0.67	19 924	1 314	347 140	262 239	0.76	4.27	4.80	29.00
Countdown	67 582	42 914	0.63	14 759	1 140	282 910	221 114	0.78	4.19	5.15	29.81
Parts	20 254 455	14 298 995	0.71	1 331 971	108 940	77 238 658	62 811 959	0.81	3.81	4.39	4.15
Whilex	228 910	98 883	0.43	47 613	123	807 645	700 703	0.87	3.53	7.09	19.34
Clausify	565 127	181 571	0.32	92 060	3 423	1 865 500	1 577 713	0.85	3.30	8.69	12.64
Queens	206 259	156 086	0.76	48 190	5 307	575 472	486 475	0.85	2.79	3.12	19.86
TribeLie	38 055	24 577	0.65	11 482	513	104 307	85 367	0.82	2.74	3.47	31.33
Fibonacci	9 427	6 154	0.65	1 149	429	16 709	13 739	0.82	1.77	2.23	35.62
Factorial	4 339	2 228	0.51	325	160	5 718	4 201	0.73	1.32	1.89	36.67
SumEuler	123 920	63 406	0.51	16 721	1 997	100 946	54 584	0.54	0.81	0.86	23.21
nofib-imaginary											
Paraffins	92 251	66 056	0.72	20 232	2 446	–	–	–	–	–	28.28
WheelSieve2	8 690	5 314	0.61	1 415	199	81 100	59 182	0.73	9.33	11.14	35.97
WheelSieve1	5 574	3 306	0.59	809	110	43 840	29 503	0.67	7.87	8.92	35.73
Primes	12 451	5 819	0.47	1 788	182	75 633	55 602	0.74	6.07	9.56	34.73
Kahan	18 451	9 688	0.53	1 662	483	82 300	63 982	0.78	4.46	6.60	34.74
Integrate	191 417	86 968	0.45	11 222	3 517	720 908	556 639	0.77	3.77	6.40	–
NQueens	53 041	39 385	0.74	11 991	1 133	124 867	105 770	0.85	2.35	2.69	29.43
GenRegexps	9 405 962	2 184 980	0.23	706 654	104 443	18 860 261	16 399 199	0.87	2.01	7.51	2.75
Tak	13 387	9 756	0.73	3 077	219	19 965	15 275	0.77	1.49	1.57	33.88
RFib	47 918	21 078	0.44	2 151	572	39 484	23 762	0.60	0.82	1.13	30.06
X2n1	86 576	36 844	0.43	4 298	2 208	48 760	25 388	0.52	0.56	0.69	26.15
nofib-spectral											
Atom	15 906	9 271	0.58	1 784	250	–	–	–	–	–	–
Life	24 018	17 185	0.72	5 855	463	267 312	217 716	0.81	11.13	12.67	–
Constraints	213 600	153 491	0.72	58 336	2 900	2 302 187	1 875 718	0.81	10.78	12.22	23.46
Cse	68 514	47 010	0.69	15 711	1 045	717 091	584 623	0.82	10.47	12.44	29.72
Lcss	55 276	45 663	0.83	8 888	332	454 307	369 461	0.81	8.22	8.09	–
Ansi	233 123	127 465	0.55	36 514	2 384	1 905 310	1 594 433	0.84	8.17	12.51	–
Calendar	1 232 767	487 357	0.40	174 031	13 530	10 004 626	8 238 879	0.82	8.12	16.91	–
Scc	16 289	11 388	0.70	4 122	103	114 698	91 428	0.80	7.04	8.03	35.17
Awards	1 443 731	681 289	0.47	225 844	21 145	10 148 588	8 181 794	0.81	7.03	12.01	–
Multiplier	3 896 127	2 008 814	0.52	558 207	25 938	25 594 685	21 202 391	0.83	6.57	10.55	–
PrettyN	19 389	9 969	0.51	3 426	272	114 615	87 248	0.76	5.91	8.75	34.74

Program	MicroHs (RV32-xfun)					GHC (RV32)			GHC/MicroHs		
	cyc	instret	IPC	combi	fn.y	cyc	instret	IPC	cyc	instret	whole
Fish	2 950 673	1 062 583	0.36	310 733	53 643	16 457 961	13 571 170	0.82	5.58	12.77	7.65
Sorting	51 202	34 135	0.67	10 110	330	278 299	221 038	0.79	5.44	6.48	30.49
Minimax	5 825 338	2 732 263	0.47	920 934	51 823	27 242 534	23 274 663	0.85	4.68	8.52	–
Cichelli	688 113	558 300	0.81	117 843	8 186	3 048 618	2 641 114	0.87	4.43	4.73	–
Lambda	113 851	81 234	0.71	31 140	761	431 833	366 292	0.85	3.79	4.51	24.43
Grep	8 987 266	3 086 321	0.34	1 039 603	73 953	24 272 685	20 390 923	0.84	2.70	6.61	–
Cryptarithm1	2 525 294	1 103 980	0.44	304 118	68 281	6 518 309	5 581 419	0.86	2.58	5.06	–
Mandel	394 335	169 825	0.43	19 417	6 926	984 597	815 349	0.83	2.50	4.80	14.33
LastPiece	1 290 419	693 983	0.54	177 979	10 049	3 088 574	2 620 849	0.85	2.39	3.78	–
Knights	11 507 455	6 739 535	0.59	1 834 309	108 095	19 776 993	17 083 661	0.86	1.72	2.53	–
Sphere	947 908	409 729	0.43	63 614	15 289	1 506 832	1 171 404	0.78	1.59	2.86	8.04
Treejoin	9 754 215	4 015 715	0.41	393 636	16 739	–	–	–	–	–	1.72
Mandel2	1 306 881	688 565	0.53	108 985	25 200	423 131	233 854	0.55	0.32	0.34	5.37
nofib-real											
Infer	3 425 601	1 623 776	0.47	587 696	10 921	13 900 962	11 297 129	0.81	4.06	6.96	–
Prolog	2 892 573	1 215 716	0.42	488 928	17 570	–	–	–	–	–	6.25
Parser	3 374 090	1 350 785	0.40	504 990	4 908	4 293 325	3 387 281	0.79	1.27	2.51	3.32
Compress	30 696 709	12 992 239	0.42	3 348 657	10 371	30 619 690	26 514 548	0.87	1.00	2.04	–
nofib-shootout											
NBody	3 595 635	2 047 290	0.57	240 404	33 911	26 381 366	21 523 172	0.82	7.34	10.51	8.85
Fft	207 393	112 843	0.54	9 480	2 225	1 092 785	924 601	0.85	5.27	8.19	21.05
SpecNorm	1 151 262	514 536	0.45	64 591	26 343	2 667 349	2 049 845	0.77	2.32	3.98	–
nofib-hartel											
Wang	138 325	64 129	0.46	16 087	1 913	813 684	645 608	0.79	5.88	10.07	24.37
Typecheck	22 538 279	13 793 125	0.61	3 196 890	171 699	130 178 479	105 205 480	0.81	5.78	7.63	6.05
Event	791 358	446 175	0.56	151 924	9 911	2 918 611	2 468 005	0.85	3.69	5.53	10.78
Sched	12 466 176	4 949 458	0.40	1 492 684	7 845	–	–	–	–	–	1.58
gadt											
RoseTree	33 459	23 270	0.70	6 125	1 429	492 759	418 951	0.85	14.73	18.00	33.49
Queue	97 201	63 094	0.65	18 740	2 472	–	–	–	–	–	28.72
MergeSort	250 912	174 331	0.69	45 191	4 987	2 893 846	2 362 773	0.82	11.53	13.55	–
InsertSort	233 133	157 503	0.68	42 269	1 918	2 675 734	2 288 109	0.86	11.48	14.53	22.84
BTree	83 230	59 342	0.71	17 353	1 617	946 727	799 381	0.84	11.37	13.47	28.63
Zipper	59 105	42 431	0.72	13 197	1 290	648 926	534 251	0.82	10.98	12.59	31.01
HeapSort	193 873	121 884	0.63	33 949	2 845	–	–	–	–	–	–
QuadTree	343 117	188 715	0.55	57 557	6 768	2 509 122	2 112 789	0.84	7.31	11.20	18.42
Patricia	101 047	74 244	0.73	15 923	3 535	600 964	508 469	0.85	5.95	6.85	26.68
List	534	139	0.26	2	0	–	–	–	–	–	38.36
Tuple	9 379	4 199	0.45	821	297	1 205	756	0.63	0.13	0.18	35.49
nofib-gc											
Power	2 631 871	1 031 446	0.39	321 235	15 764	5 177 280	4 256 151	0.82	1.97	4.13	4.57
Circsim	18 610 231	10 959 739	0.59	1 646 158	67 330	32 290 938	26 666 641	0.83	1.74	2.43	–

TABLE II: Cache behaviour. MicroHs counters are for the compute window. The GHC counters are emitted once per run by the harness and therefore cover the **whole run**, RTS start-up included; they are not directly comparable with the MicroHs columns and no ratio is given.

Program	MicroHs (RV32-xfun), compute window						GHC (RV32), whole run			
	i-miss	d-miss	fills	miss stall	RAM rd	RAM wr	i-miss	d-miss	fills	miss stall
flite										
Permsort	327	1	20	706	416	1 994	326	1 129	1 418	25 721
Queens2	442	1	27	906	576	9 791	393	1 100	1 473	27 088
Taut	464	2	32	1 064	640	48 306	408	1 096	1 482	27 033
Ordlist	356	1	17	693	368	537	268	510	770	13 950
Braun	1 168	1	291	8 302	4 816	107 823	430	2 524	2 922	53 080
Sumpuz	293	1	50	1 208	945	5	454	219	620	10 942
Sudoku	12 831	20	6 009	190 805	97 936	1 050 625	2 254	14 572	16 500	306 253

Program	MicroHs (RV32-xfun), compute window						GHC (RV32), whole run			
	i-miss	d-miss	fills	miss stall	RAM rd	RAM wr	i-miss	d-miss	fills	miss stall
TreeSum	4 470	4	18	4 904	480	457 850	136	4 704	4 829	87 286
TreePari	982	1	12	1 227	336	48 611	88	604	684	12 233
Adjoxo	974	1	50	1 923	944	23 594	343	219	547	9 639
Mss	4 199	2	233	9 654	14 224	331 612	283	1 926	2 195	39 846
Knuthbendix	1 036	3	366	7 814	6 464	31 120	1 443	472	1 821	32 090
Countdown	999	1	53	1 939	960	16 672	455	488	899	16 231
Parts	56 381	118 077	124 697	2 565 232	2 042 272	8 626 772	2 475	87 950	90 185	1 708 622
Whilex	2 086	1	79	3 212	1 360	168 426	465	1 136	1 568	28 810
Clausify	3 775	1	61	5 022	1 104	448 085	457	1 942	2 384	43 655
Queens	3 241	1	29	3 853	560	129 278	148	582	723	12 967
TribeLie	752	1	27	1 310	2 384	4 222	146	222	360	6 342
Fibonacci	48	1	9	215	241	2	21	7	26	422
Factorial	7	1	8	160	609	226	19	15	34	604
SumEuler	416	1	21	788	624	39 157	40	11	50	823
nofib-imaginary										
Paraffins	533	1	58	1 549	1 600	35 833	956	1 438	2 335	42 596
WheelSieve2	120	1	47	951	849	3	576	151	696	12 288
WheelSieve1	78	1	41	820	753	3	480	91	555	9 911
Primes	100	1	18	400	401	2	136	103	227	4 100
Kahan	187	1	31	707	785	41	528	168	641	11 300
Integrate	701	1	38	1 365	833	51 983	848	817	1 589	28 646
NQueens	566	1	23	968	448	7 211	109	68	168	3 113
GenRegexps	218 063	627	98 315	2 540 312	2 372 881	5 503 842	863	31 375	32 150	602 049
Tak	120	1	11	324	401	3	21	6	26	495
RFib	241	1	16	495	433	2	361	65	384	6 691
X2n1	386	1	46	1 185	1 377	396	563	110	621	10 721
nofib-spectral										
Atom	64	1	32	622	961	198	847	1 500	2 300	41 345
Life	207	2	54	1 158	1 345	244	485	396	868	15 563
Constraints	1 651	3	88	3 316	1 600	139 433	1 137	3 352	4 428	79 683
Cse	623	1	53	1 528	4 272	19 035	626	793	1 383	25 056
Lcss	209	1	45	1 027	913	15	539	721	1 205	21 648
Ansi	2 594	1	175	5 868	24 048	140 255	713	4 070	4 747	85 965
Calendar	6 590	2	4 059	137 381	110 608	901 229	1 625	16 427	17 933	334 598
Scc	234	1	48	1 068	1 089	71	776	207	933	16 525
Awards	10 002	40	538	25 600	18 944	1 159 776	1 449	12 593	13 896	256 903
Multiplier	32 796	28	13 292	409 301	293 985	2 915 285	3 255	30 560	33 576	631 878
PrettyN	297	1	113	2 345	3 441	1 388	950	298	1 172	20 705
Fish	29 211	4	16 163	486 105	374 096	2 033 858	4 282	26 810	30 876	580 848
Sorting	477	1	147	3 161	4 144	6 943	1 251	494	1 654	29 220
Minimax	33 609	3	1 895	89 277	34 160	4 831 365	3 549	32 663	36 028	681 275
Cichelli	3 371	1	403	14 263	10 224	496 904	899	2 355	3 186	57 377
Lambda	1 562	1	88	3 441	1 632	60 276	418	453	836	15 098
Grep	103 925	1	42 143	1 322 279	1 207 056	6 410 635	1 739	29 364	30 934	579 687
Cryptarithm1	19 479	1	123	21 693	2 096	2 087 257	686	9 697	10 332	188 982
Mandel	1 417	1	55	2 369	4 368	113 926	907	888	1 714	30 626
LastPiece	6 159	4	1 159	40 235	32 400	1 008 166	1 166	2 933	4 040	73 466
Knights	55 071	4 949	5 089	182 079	87 824	9 120 952	2 803	10 215	12 785	236 851
Sphere	4 886	6	587	16 989	9 904	383 256	2 842	1 173	3 747	66 887
Treejoin	209 156	82	228 798	4 688 625	3 684 736	2 330 492	1 209	15 340	16 460	303 795
Mandel2	9 986	4	199	14 587	3 328	642 405	607	225	758	13 300
nofib-real										
Infer	18 949	19	3 931	136 627	76 432	2 795 880	5 011	11 433	16 083	297 136
Prolog	20 251	36	5 470	196 825	121 025	2 289 075	4 555	13 460	17 749	331 800
Parser	32 262	3	4 066	157 419	93 856	2 749 978	2 829	4 950	7 615	137 790
Compress	181 052	32 027	42 249	1 148 886	843 712	20 820 292	1 504	34 710	36 098	677 245
nofib-shootout										
NBody	66 271	10	22 626	525 408	369 569	1 900 436	8 496	61 568	69 474	1 320 633
Fft	724	12	89	2 271	2 065	52 487	1 273	1 672	2 871	51 635
SpecNorm	12 476	620	131	20 162	60 672	530 655	1 175	3 902	4 994	90 994
nofib-hartel										

Program	MicroHs (RV32-xfun), compute window						GHC (RV32), whole run			
	i-miss	d-miss	fills	miss stall	RAM rd	RAM wr	i-miss	d-miss	fills	miss stall
Wang	869	4	117	3 110	4 912	48 017	1 624	1 089	2 610	46 839
Typecheck	95 900	24 902	47 698	1 310 044	878 080	15 502 284	14 939	88 679	102 601	1 952 934
Event	2 215	7	358	12 438	8 672	613 124	1 142	3 274	4 360	78 084
Sched	108 703	13 230	71 101	1 872 979	1 138 048	8 508 749	2 048	14 729	16 619	308 082
gadt										
RoseTree	208	1	36	863	1 793	962	615	575	1 124	19 710
Queue	492	1	38	1 144	6 816	40 088	875	1 652	2 447	44 468
MergeSort	2 216	1	39	2 975	8 144	155 571	438	3 221	3 615	65 386
InsertSort	497	1	29	974	14 736	137 802	283	3 953	4 209	75 721
BTree	599	1	44	1 380	2 048	33 307	530	1 329	1 825	33 763
Zipper	533	2	49	1 398	2 000	13 578	755	987	1 666	30 087
HeapSort	593	2	34	1 178	14 144	111 670	771	3 299	3 999	72 510
QuadTree	3 112	4	117	5 294	5 584	252 932	1 207	2 694	3 787	68 356
Patricia	1 105	2	67	2 280	2 544	40 627	704	727	1 380	24 482
List	5	1	5	114	129	1	1	3	3	58
Tuple	90	1	36	748	1 953	1 122	344	64	373	6 369
nofib-gc										
Power	68 382	13	17 494	415 565	288 192	1 804 559	1 364	6 149	7 404	134 167
Circsim	148 662	42 783	161 864	4 610 922	2 633 744	10 009 443	9 623	47 334	56 443	1 064 057

TABLE III: Full breakdown behind Table III of the letter, three-compiler execution time (compute cycles). Ratios are relative to GCC.

Program	GCC `-O3`	GHC	MicroHs	GHC/GCC	MicroHs/GCC
shootout					
nbody	948k	28.84M	5.05M	30.42	5.33
matrix	215k	25.36M	8.42M	117.95	39.16
sieve	286k	24.83M	5.85M	86.82	20.45
binarytrees	11.28M	22.06M	8.48M	1.96	0.75
fannkuchredux	99k	17.58M	4.48M	177.58	45.25
takfp	2.07M	15.99M	6.00M	7.72	2.90
methcall	1.17M	11.37M	13.54M	9.72	11.57
magicsquares	256k	8.42M	2.13M	32.89	8.32
nsievebits	40.5k	585k	311k	14.44	7.68
strcat	313k	7.17M	1.09M	22.91	3.48
objinst	6.02M	4.96M	4.80M	0.82	0.80
ary	32.1k	8.34M	2.89M	259.81	90.03
spectralnorm	242k	3.69M	1.81M	15.25	7.48
mandelbrot	2.16M	3.03M	2.61M	1.40	1.21
alloc	2.93M	2.71M	9.56M	0.92	3.26
spec.norm-pure	324k	2.40M	1.52M	7.41	4.69
nestedloop	7187	1.96M	8.71M	272.71	1211.91
heapsort	30k	1.60M	419k	53.33	13.97
hash	230k	1.56M	2.70M	6.78	11.74
wc	250k	1.20M	406k	4.80	1.62
random	767k	941k	6.92M	1.23	9.02
nsieve-pure	116k	232k	58k	2.00	0.50
nsieve	8120	115k	33k	14.16	4.06
rfib	7513	84k	68k	11.18	9.05
fasta	832k	2.63M	1.23M	3.16	1.48
harmonic	11.8k	941k	1.29M	79.75	109.32
partialsums	1.50M	6.23M	16.41M	4.15	10.94
others					
floydwarshall	24k	2.36M	1.01M	98.33	42.08
crc32	112k	1.77M	1.70M	15.80	15.18
fft	176k	1.16M	264k	6.59	1.50
heapsort (heap)	214k	509k	207k	2.38	0.97
tak	9052	20k	23k	2.21	2.54
gadt					
quadtree	372k	1.26M	348k	3.39	0.94
btree	300k	473k	84k	1.58	0.28
zipper	207k	379k	60k	1.83	0.29
patricia	136k	303k	102k	2.23	0.75
rosetree	144k	249k	34k	1.73	0.24

TABLE IV: Full PMC breakdown for the nofib programs ported and measured after the letter was written; not in Table II of the letter. All counters are one run of the current bbird-fnadd model, compute window, except *whole* which is the whole run including start-up. Every program returns the same answer as GHC. The window satisfies cycles = instret + bubble + freeze + residual on every row.

Program	issue			cycle split			stalls / branch				d-cache		misses and RAM					cycles
	cyc	instret	IPC	resid	freeze	bubble	starve	stall ld	misp T	misp NT	fill	wb	i-miss	d-miss	RAM rd	RAM wr	miss stall	whole
Banner	174 497	80 458	0.46	2 755	10 233	81 051	61 973	3 049	44	46	290	4 044	782	6	27 564	82 822	6 866	466 002
Boyer	8 898 934	3 728 818	0.42	99 632	3 977 381	1 093 103	925 291	110 975	25 594	30 012	10 479	456 953	66 616	9	168 941	7 279 739	386 416	9 204 603
Compress2	5 019 770	1 797 833	0.36	64 645	2 401 412	755 880	645 757	37 035	2 793	8 006	11 485	252 767	43 660	89	217 948	3 979 979	352 282	5 342 960
DigitsE1	435 484	237 418	0.55	8 587	111 142	78 337	51 516	13 132	810	571	144	19 058	3 573	4	3 084	304 921	6 293	741 307
Eliza	15 929 259	7 617 162	0.48	297 561	6 305 941	1 708 595	1 437 541	251 532	5 571	6 558	10 873	825 817	99 399	27	218 508	13 202 534	417 395	16 252 962
Expert	929 147	429 717	0.46	18 757	350 226	130 447	117 609	10 489	553	644	1 806	43 839	6 876	11	33 134	697 122	62 556	1 235 658
Lift	1 559 597	548 081	0.35	18 054	611 192	382 270	347 842	9 943	633	819	6 274	72 249	15 560	16	139 468	1 151 238	201 953	1 875 671
Mkhprog	5 351 685	1 941 851	0.36	54 428	755 683	2 599 723	2 154 381	73 356	241	4 091	43 541	187 076	36 388	15	1 064 380	3 344 071	1 066 091	5 668 738
ParQueens	9 355 659	5 096 161	0.54	132 359	2 893 663	1 233 476	955 210	224 874	16 140	5 380	6 062	479 687	52 246	22	97 724	7 670 858	237 351	9 662 158
Parfib	1 546 211	1 030 903	0.67	21 522	248 477	245 309	140 598	83 382	7 058	11 564	50	75 405	2 425	11	1 181	1 206 225	3 741	1 848 164
Partak	7 872 554	3 121 239	0.40	49 706	4 011 480	690 129	555 993	111 315	15 886	6 875	1 616	410 501	55 542	40	26 252	6 562 510	103 619	8 175 367
Sumeuler	8 179 226	4 152 935	0.51	65 723	277 890	3 682 678	1 980 172	255 928	13 066	31 862	1 168	364 037	19 081	7	61 901	5 863 448	51 486	8 496 897